

# **BIOL-1 Practice Test 04 — Answer Key**

## **Comprehensive Final Review (Modules 01–16)**

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## **Part A: Multiple Choice Answers**

| Q  | Answer |
|----|--------|
| 1  | C      |
| 2  | B      |
| 3  | C      |
| 4  | B      |
| 5  | A      |
| 6  | C      |
| 7  | C      |
| 8  | D      |
| 9  | C      |
| 10 | B      |
| 11 | A      |
| 12 | D      |
| 13 | D      |
| 14 | B      |
| 15 | A      |
| 16 | A      |
| 17 | C      |
| 18 | B      |
| 19 | A      |

| <b>Q</b> | <b>Answer</b> |
|----------|---------------|
| 20       | A             |
| 21       | B             |
| 22       | D             |
| 23       | B             |
| 24       | B             |
| 25       | C             |
| 26       | B             |
| 27       | A             |
| 28       | B             |
| 29       | C             |
| 30       | C             |
| 31       | A             |
| 32       | D             |
| 33       | A             |
| 34       | C             |
| 35       | B             |
| 36       | C             |
| 37       | D             |
| 38       | D             |

| Q  | Answer |
|----|--------|
| 39 | B      |
| 40 | A      |
| 41 | B      |
| 42 | D      |
| 43 | A      |
| 44 | B      |
| 45 | B      |
| 46 | D      |
| 47 | C      |
| 48 | D      |
| 49 | C      |
| 50 | B      |
| 51 | A      |
| 52 | A      |
| 53 | C      |
| 54 | A      |
| 55 | D      |
| 56 | D      |
| 57 | C      |

| Q  | Answer |
|----|--------|
| 58 | C      |
| 59 | A      |
| 60 | C      |
| 61 | D      |
| 62 | A      |
| 63 | C      |
| 64 | D      |
| 65 | D      |
| 66 | A      |
| 67 | D      |
| 68 | B      |
| 69 | A      |
| 70 | B      |
| 71 | C      |
| 72 | D      |
| 73 | A      |
| 74 | B      |
| 75 | D      |

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## **Part B: Fill in the Blank Answers**

| Q  | Acceptable answers (synonyms in parentheses)                                                                      |
|----|-------------------------------------------------------------------------------------------------------------------|
| 76 | Control (control group)                                                                                           |
| 77 | Homeostasis                                                                                                       |
| 78 | Electrons (valence electrons / electron pairs — accept “electrons”)                                               |
| 79 | Electrons                                                                                                         |
| 80 | Peptide                                                                                                           |
| 81 | Lipids (fats / oils / triglycerides — “lipid” alone OK)                                                           |
| 82 | Lysosomes                                                                                                         |
| 83 | Nucleus                                                                                                           |
| 84 | Facilitated (facilitated diffusion; not “active”)                                                                 |
| 85 | Hydrophobic                                                                                                       |
| 86 | Glycolysis                                                                                                        |
| 87 | Chlorophyll                                                                                                       |
| 88 | DNA (DNA polymerase)                                                                                              |
| 89 | Codon                                                                                                             |
| 90 | Anaphase                                                                                                          |
| 91 | Prophase                                                                                                          |
| 92 | Phenotype                                                                                                         |
| 93 | Codominance (incomplete dominance is a different pattern — prefer codominance for “both alleles visible” framing) |

| Q   | Acceptable answers (synonyms in parentheses)                        |
|-----|---------------------------------------------------------------------|
| 94  | Epigenetic                                                          |
| 95  | Factors (transcription factors)                                     |
| 96  | Cloning (recombinant DNA / genetic engineering / molecular cloning) |
| 97  | Ligase (DNA ligase)                                                 |
| 98  | Variation (heritable variation / genetic variation)                 |
| 99  | Convergent                                                          |
| 100 | Drift (genetic drift)                                               |
| 101 | 2pq (2 pq)                                                          |
| 102 | Prezygotic                                                          |
| 103 | Postzygotic                                                         |
| 104 | Population                                                          |
| 105 | Producers (primary producers; autotrophs)                           |

## Part C: Short Answer / Free Response Key (32 questions)

Scoring: accept reasonable wording; require the **bold ideas** in each rubric for full credit unless a solid alternative demonstrates the same concept.

**106. (Module 01) Two characteristics of life:** Any **two** from: **cellular** organization, **reproduction**, **metabolism**, **response** to stimuli, **homeostasis**, **growth/development**, **heritable change** in populations over time / evolution (definitions should match the textbook sense).

**107. (Module 01) Theory vs hypothesis:** **Hypothesis** = one testable explanation; **theory** = broad explanatory framework supported by **many** lines of evidence (not “just a guess” in science).

**108. (Module 02) Polar water dissolves salt:** Water is **polar** partial charges **attract ions** pulled off/away from the crystal (hydration shells / dissociation concept).

**109. (Module 02) Buffer: Resists pH change** by taking up or releasing  $H^+$  / binds acid or base (bicarbonate buffer in blood is a classic example).

**110. (Module 03) Four macromolecules + monomer:** **Carbohydrates, lipids, proteins, nucleic acids** — monomer examples: **monosaccharide, glycerol+fatty acids** (or fatty acid), **amino acid, nucleotide**.

**111. (Module 03) Denatured protein: Loses shape → often loses function;** cause: **heat, pH** extremes, harsh chemicals, etc.

**112. (Module 04) Prokaryote vs eukaryote:** Prokaryotes **no membrane nucleus**, smaller/less compartmentalized; eukaryotes **membrane-bound nucleus** and **organelles** (two clear contrasts).

**113. (Module 04) Mitochondria: ATP / energy** from **nutrients**; work done mainly in **mitochondrial** compartments (matrix / electron transport / oxidative phosphorylation — intro wording OK).

**114. (Module 05) Passive vs active:** **Passive** = with gradient, no ATP for the step (e.g. **diffusion, facilitated diffusion**); **Active** = **against** gradient, **uses ATP** (e.g. **sodium–potassium pump**).

**115. (Module 05) Very salty outside, water-permeable animal cell:** Water moves **out** → **crenation / shrinkage** (hypertonic outside — framing should match water leaving).

**116. (Module 06) Photosynthesis & respiration:** Plants fix **carbon/energy** in sugars; respiration **releases** stored energy; **matter** links as **CO<sub>2</sub> / glucose / O<sub>2</sub>** in the big cycle (complementary, not “same process”).

**117. (Module 06) Enzyme high temperature: Denatures / unfolds** the enzyme → **loses** proper **active site** shape → **slows** or **stops** catalysis.

**118. (Module 07) Transcription vs translation: Transcription: DNA → RNA** (nuclear in euk); **Translation: mRNA → protein** at **ribosome** (codon/amino acid link implied).

**119. (Module 07) Semiconservative replication:** Each new double helix has **one parent** strand and **one new** strand (template-based copying).

**120. (Module 08) Mitosis vs meiosis goals: Mitosis: two** genetically **identical** diploid daughter cells for **growth/repair**; **Meiosis: four haploid** gametes with **variation** for **sexual reproduction**.

**121. (Module 08) Meiosis variation: Crossing over; independent assortment;** (also **random fertilization** if not used elsewhere).

**122. (Module 09)  $Aa \times Aa$  phenotype: 3 dominant : 1 recessive** (classic **3:1** phenotypes if complete dominance).

**123. (Module 09) Heterozygous: Two different alleles;** example **Aa, Bb**, etc.

**124. (Module 10) Epigenetics vs mutation: Mutation = DNA sequence** change; **epigenetics = expression** change **without** changing the **sequence** (often reversible “dials”).

**125. (Module 10) Environment + trait expression:** Example: **nutrition** affecting growth, **stress** / chemicals influencing methylation patterns, **temperature** affecting enzyme activity in **regulation** contexts — must tie to **gene activity without** claiming offspring inherit **changed DNA letters** from gym workouts.

**126. (Module 11) GMO + ethics: Foreign gene / engineered** organism or crop; ethics: **food safety, labeling, environment, ownership, unintended effects** (any two sensible tensions).

**127. (Module 11) Gel electrophoresis: Smaller** fragments move **farther** / **faster** in the gel than **larger** ones (separation by **size**).

**128. (Module 12) Natural selection (friend explanation):** Variation is **heritable**; **more offspring** than can survive; **environment** “sorts” who reproduces; **favorable** traits become **more common over generations** (four-sentence flow OK).

**129. (Module 12) Two evidence lines:** e.g. **fossils** (historical change), **homologous** structures (**common ancestry**), **biogeography** (island patterns), **DNA** similarities — each with **one** clause of what it shows.

**130. (Module 13) Microevolution + non-selection mechanism:** **Allele frequency** change in a **population**; **genetic drift**, **gene flow**, **mutation** (plus selection — but question asked **other than** selection).

**131. (Module 13) Selection modes:** **Directional** favors **one extreme**; **stabilizing** favors **intermediate**; **disruptive** favors **both extremes**.

**132. (Module 14) Allopatric vs sympatric:** **Allopatric:** **geographic** barrier then divergence; **Sympatric:** **same region** (e.g. **polyploidy**) new species.

**133. (Module 14) Barrier examples:** **Prezygotic:** e.g. **different mating season, behavior, habitat split**; **Postzygotic:** **hybrid sterile, inviable zygote**.

**134. (Module 15) Exponential vs logistic:** **Exponential:** rapid **unlimited-phase** growth; **logistic:** growth **slows** as it approaches **carrying capacity K** (resources/limits).

**135. (Module 15) 10% rule & chains:** **~10%** energy **transferred** to **next trophic level** (rest **heat / metabolism**); **more links** → **more loss** → **fewer** top predators supported from the same **producer base**.

**136. (Module 16) Cross-scale system:** A complete response links **genetic information/variation** to a trait or process, explains how **selection** can change trait frequency over generations, and ties the result to **resource limits**, trophic interactions, carrying capacity, or nutrient/energy constraints.

**137. (Module 16) Evidence scale:** Strong responses state the scale because different evidence answers different questions: molecules show mechanisms, cells show processes, organisms show traits, populations show allele or abundance change, communities show interactions, and ecosystems show energy/matter constraints.

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*End of Answer Key*